Decent Coaching Classes

VII SB

Subject: Mathematics — ANSWER KEY

Chapters: 1. Geometrical Constructions   2. Multiplication and Division of Integers   3. HCF and LCM

Q.1 (A) Choose the correct alternative. **[4]**

1. (b) incentre
2. (b) always positive
3. (c) co-prime numbers
4. (a) 72   [LCM = Product ÷ HCF = 288 ÷ 4 = 72]

Q.1 (B) Fill in the blanks. **[4]**

1. congruent
2. 72
3. product
4. co-prime (relatively prime)

Q.2 (any four) **[8]**

1. Draw seg BC. At point B, draw a ray making an angle of 70° with BC using a protractor to get ∠ABC = 70°. With the compass at the required opening, mark equal arcs on both arms from B and then from those points to get intersection point; draw ray BM through this point — ray BM is the bisector, dividing ∠ABC into two angles of 35° each.
2. $(-15) \times 12 = -180$
3. $(-144) \div 12 = -12$
4. $36 = 2 \times 2 \times 3 \times 3$; $48 = 2 \times 2 \times 2 \times 2 \times 3$ $\Rightarrow$ HCF $= 2 \times 2 \times 3 = 12$
5. $84 = 2 \times 2 \times 3 \times 7$
6. Any two pairs, e.g. (3, 5), (5, 7), (11, 13), (17, 19)

Q.3 (any three) **[9]**

1. Rough figure: draw seg AB = 5 cm as base. With A as centre and radius 4.5 cm, draw an arc. With B as centre and radius 6 cm, draw another arc cutting the first at C. Join AC and BC to get the required ΔABC.
2. $(-125) \div (-25) = 5$; $5 \times (-4) = -20$
3. $24 = 2^3 \times 3$; $36 = 2^2 \times 3^2$; $40 = 2^3 \times 5$ $\Rightarrow$ LCM $= 2^3 \times 3^2 \times 5 = 360$
4. $126 = 84 \times 1 + 42$; $84 = 42 \times 2 + 0$ $\Rightarrow$ HCF $= 42$

Q.4 (any one) **[5]**

1. Draw seg PQ = 5.5 cm as base. At P, draw ray making ∠P = 60°. With P as centre and radius 5 cm, cut an arc on this ray at R. Join QR to obtain ΔPQR. Then draw the perpendicular bisector of each side (PQ, QR, RP) using a compass — the three bisectors meet at one point, the circumcentre, equidistant from all three vertices.
2. $168 = 2^3 \times 3 \times 7$; $210 = 2 \times 3 \times 5 \times 7$; $294 = 2 \times 3 \times 7^2$ $\Rightarrow$ HCF $= 2 \times 3 \times 7 = 42$ m (greatest possible length).

   Number of pieces: $168 \div 42 = 4$; $210 \div 42 = 5$; $294 \div 42 = 7$